Fishing for Sustainability

Assignment Submission

Wed Sep 9, 2026

Name:____________________________________

Each person should submit their own copy of this handout at the end of class for today's completion credit.

Round 1: Open Access

Data Table

Item	Value
Your catch	
Total class catch	
Fish remaining after harvest	
Fish after reproduction (doubled, max 30)	

Reflection

(1) Based on what happened, is this situation a **social dilemma**? Explain briefly.

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(2) In one sentence, explain how Round 1 illustrates the **Tragedy of the Commons** (focus on individual incentives vs. group outcomes).

Round 2: Fixed Quota

Data Table

Item	Value
Agreed class quota	
Your catch	
Total class catch	
Fish remaining after harvest	
Fish after reproduction (doubled, max 30)	

Reflection

(3) During the discussion, did the class choose a quota aimed at **maximizing long-term sustainability** or **maximizing short-term catch**?

(4) How does this round illustrate challenges of **enforcement, trust, or cheating** in managing common-pool resources?

Round 3: Fixed Effort

Data Table

Item	Value
Agreed per-person limit	
Total class catch (limit × 8)	
Fish remaining after harvest	
Fish after reproduction (doubled, max 30)	

Reflection

(5) Was it easier to comply with this rule compared to the quota in Round 2? Why or why not?

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(6) How does this round reduce (or fail to reduce) the **social dilemma** you experienced earlier?

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Synthesis

Data Overview

Round	Strategy	Total.Catch	Fish.Remaining.After.Harvest	Fish.After.Reproduction
1	Open Access			
2	Fixed Quota			
3	Fixed Effort			

Synthesis Questions

(7) Looking at your data, which round left the fish population **closest to the midpoint (~15 fish)** after harvest? Why does leaving the population near this level tend to produce the **highest sustainable yield over time**?

(8) For each strategy, give **one advantage and one drawback** based on your experience:

- Open Access:

- Fixed Quota:

- Fixed Effort (Per-Person Limit):

(9) Which round depended most on **trust** between individuals? Which round relied most on **enforcement**? Why are these factors so important in managing real-world common-pool resources?

(10) If you were managing a real fishery, which strategy (or combination of strategies) would you recommend?

(11) Name **two specific actions** you would implement in your strategy from the previous question (e.g., monitoring, enforcement, adaptive limits, stakeholder involvement).

(12) Give one example of another **common-pool resource** (e.g., groundwater, forests, wildlife, carbon emissions). Which management approach from this activity might work best there—and why?